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Equalizer bearing assembly having wedged retainers

Abstract

An assembly is used to connect an equalizer bar to a walking beam of a reciprocating pump unit. The walking beam has a bearing shaft having cylindrical ends. Brackets attached to the equalizer bar define slots, each having a hooked section and having a base section. The hooked sections are configured to rest against the cylindrical ends of the bearing shaft. The base sections of the slots oppose the hooked sections at a gap from the cylindrical ends of the bearing shaft. Wedges are configured to position in the slots between the bases and the cylindrical ends of the bearing shaft.

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Background/Summary

BACKGROUND OF THE DISCLOSURE

(1) Reciprocating pump systems, such as sucker rod pump systems, extract fluids from a well and employ a downhole pump connected to a driving source at the surface. A rod string connects the surface driving force to the downhole pump in the well. When operated, the driving source cyclically raises and lowers the downhole pump, and with each stroke, the downhole pump lifts

well fluids toward the surface.

(2) For example, FIG. **1** shows a sucker rod pump system **10** used to produce fluid from a well. A downhole pump **14** has a barrel **16** with a standing valve **24** located at the bottom. The standing valve **24** allows fluid to enter from the wellbore, but the standing valve **24** does not allow the fluid to leave. Inside the pump barrel **16**, a plunger **20** has a traveling valve **22** located at the top. The traveling valve **22** allows fluid to move from below the plunger **20** to the production tubing **18** above, but the traveling valve **22** does not allow fluid to return from the tubing **18** to the pump barrel **16** below the plunger **20**. A driving source (e.g., a pump jack or pumping unit **30**) at the surface connects by a rod string **12** to the plunger **20** and moves the plunger **20** up and down cyclically in upstrokes and downstrokes.

(3) During the upstroke, the traveling valve **22** is closed, and any fluid above the plunger **20** in the production tubing **18** is lifted towards the surface. Meanwhile, the standing valve **24** opens and allows fluid to enter the pump barrel **16** from the wellbore.

(4) At the top of stroke, the standing valve **24** closes and holds in the fluid that has entered the pump barrel **16**. Furthermore, throughout the upstroke, the weight of the fluid in the production tubing **18** is supported by the traveling valve **22** in the plunger **20** and, therefore, also by the rod string **12**, which causes the rod string **12** to stretch. During the downstroke, the traveling valve opens, which results in a rapid decrease in the load on the rod string **12**. The movement of the plunger **20** from a transfer point to the bottom of stroke is known as the “fluid stroke” and is a measure of the amount of fluid lifted by the pump **14** on each stroke.

(5) At the surface, the pump unit **30** is driven by a prime mover **40**, such as an electric motor or internal combustion engine, mounted on a pedestal above a base **32**. Typically, a pump controller **36** monitors, controls, and records the pump unit's operation. Structurally, a Samson post **34** on the base **32** provides a fulcrum on which a walking beam **50** is pivotally supported by a saddle bearing assembly **35**.

(6) Output from the motor **40** is transmitted to a gearbox **42**, which provides low-speed, high-torque rotation of a crankshaft **43**. Both ends of the crankshaft **43** rotate crank arms **44** having counterbalance weights **46**. Each crank arm **44** is pivotally connected to a pitman arm **48** by a crank pin bearing **45**. In turn, the two pitman arms **48** are connected to an equalizer bar **49**, which is pivotally connected to the rear end of the walking beam **50** by an equalizer bearing assembly **55**.

(7) A horsehead **52** with an arcuate forward face **54** is mounted to the forward end of the walking beam **50**. As is typical, the face **54** may have tracks or grooves for carrying a flexible wire rope bridle **56**. At its lower end, the bridle **56** terminates with a carrier bar **58**, upon which a polished rod **15** is suspended. The polished rod **15** extends through a packing gland or stuffing box at the wellhead **13**. The rod string **12** of sucker rods hangs from the polished rod **15** within the tubing string **18** located within the well casing and extends to the downhole pump **14**.

(8) A hook-type attachment has been used for decades on the equalizer bearing assembly **55** to connect the equalizer beam **49** to a bearing shaft on the beam **50**. FIGS. **2**A-**2**C show features of a hook-type attachment **55***a* according to the prior art. As shown in FIG. **2**A, the attachment **55***a* includes an equalizer shaft **60**, a bearing **62**, a bearing housing **64**, and a retaining ring **66**. As shown in FIG. **2**B, the equalizer shaft **60** fits in the bearing **62**, and bearing housing **64** supports the bearing **62** by clamping thereabout with fasteners **65**. The bearing housing **64** is affixed to the walking beam **50**. The equalizer bar **49** has hooks **49***h* that fit onto the free ends of the equalizer shaft **60** so the equalizer bar **49** can be supported on the walking beam **50** and can articulate relative thereto. As shown in the detail of FIG. **2**C, retaining blocks **70** can be affixed by bolts **72** to the hooks **49***h* so the retaining blocks **70** rest under free ends of the equalizer shaft **60**.

(9) As will be appreciated, this hook-type attachment **55***a* in FIGS. **2**A-**2**C is designed to only withstand tensile loading. The retaining blocks **70** and bolts **72** only retain the hook **49***h* to the equalizer shaft **60** under ideal operating conditions. If there is compressive loading (e.g., in the case of a polished rod hang-up, high rod part, or other type of abrupt shock loading), the bolts **72** for the

retaining blocks **70** experience large amounts of bending stresses, which can often result in failure of this connection and produce catastrophic collapse of the structure.
(10) Other than a hook-type attachment as discussed above, bolted clamps have also been used in the past to retain an equalizer beam to an equalizer bearing shaft. FIGS. **3**A-**3**B show a bolted clamp attachment **55***b* used to attach an equalizer bar **49** to an equalizer shaft **60**. The attachment **55***b* includes an equalizer shaft **60**, a bearing **62**, and a bearing housing **64**. The bearing housing **64** bolts to the underside of the walking beam (not shown). Clamps **68** are affixed to the equalizer bar **49** and clamp around the free ends of the equalizer shaft **60**. Bolts hold the ends of the clamps **68** together.
(11) Assembling this bolted clamp attachment **55***b* is time consuming. If the bolts are not torqued properly upon initial installation and are not inspected and re-tightened periodically, the bolts are prone to fatigue and can fail prematurely causing the catastrophic collapse of the pumping unit.
(12) The subject matter of the present disclosure is directed to overcoming, or at least reducing the effects of, one or more of the problems set forth above.
SUMMARY OF THE DISCLOSURE
(13) An assembly disclosed herein is used to connect arms of a reciprocating pump unit to a walking beam of the reciprocating pump unit. The walking beam has a bearing shaft with cylindrical ends. The assembly comprises an equalizer bar and wedges. The equalizer bar has brackets, and each of the brackets defines a slot. The slot has a hooked section and has a base section. The hooked section is configured to rest against a respective one of the cylindrical ends of the bearing shaft, and the base section opposes a respective one of the hooked sections at a gap from the cylindrical end of the bearing shaft. The wedges are configured to position in the slots between the base sections and the cylindrical ends of the bearing shaft. Each of the wedges has first and second edges. The first edge is configured to rest against the cylindrical end of the bearing shaft, and the second edge opposes the first edge and is configured to rest against the base section of the slot.
(14) The assembly can comprise fasteners or bolts, each affixing a respective one of the wedges in a respective one of the slots. Each of the wedges can define an opening therethrough, and each of the slots can define a bolt hole. Each of the bolts can be configured to position in the opening defined through the wedge and to thread into the bolt hole in the slot.
(15) The first edge can have a concave profile, and the second edge can have a convex profile. The concave profile can define a first segment that is larger than a second segment of the convex profile. The concave profile can define a first radius of curvature matching the cylindrical end of the bearing shaft. The convex profile can define a second radius of curvature at a central axis offset from a rotational axis of the bearing shaft. The base section can define a corresponding concave profile being complementary to the convex profile of the second edge. The convex profile can define a radius of curvature at a central axis offset from a rotational axis of the bearing shaft.
(16) The assembly can comprise a bearing housing configured to attach to the walking beam. The bearing housing has a bearing configured to support the bearing shaft for rotation therein.
(17) An assembly disclosed herein is configured to connect an equalizer bar to a walking beam of a reciprocating pump unit. The walking beam has a bearing shaft with cylindrical ends. The assembly comprises brackets and wedges. The brackets are configured to attach to the equalizer bar. Each of the brackets defines a slot, which has a hooked section and a base section. The hooked section is configured to rest against a respective one of the cylindrical ends of the bearing shaft, and the base section opposes a respective one of the hooked sections at a gap from the cylindrical end of the bearing shaft. The wedges are configured to position in the slots between the bases and the cylindrical ends of the bearing shaft. Each of the wedges has first and second edges. The first edge is configured to rest against the cylindrical end of the bearing shaft, and the second edge opposes the first edge and is configured to rest against the base section of the slot.
(18) A reciprocating pump unit disclosed herein comprises arms, an equalizer bar, a walking beam,

and an assembly. The arms are configured to translate on the reciprocating pump unit, and the equalizer bar is hingedly connected to the arms. The walking beam is mounted to pivot on the reciprocating pump unit, and the assembly connects the equalizer bar to the walking beam.
(19) The assembly comprises a shaft, a bearing, brackets, and wedges. The shaft has cylindrical ends, and the bearing is disposed on the walking beam and supports the shaft to rotate. The brackets are disposed on the equalizer bar, and each of the brackets defining a slot, which has a hooked section and a base section. The hooked section is configured to rest against a respective one of the cylindrical ends of the shaft, and the base section opposes a respective one of the hooked sections at a gap from the cylindrical end of the pin. The wedges are disposed in the slots between the cylindrical ends of the shaft and the base sections of the slots. Each of the wedges has first and second edges. The first edge rests against the cylindrical end of the shaft, and the second edge opposes the first edge and rests against the base section of the slot.
(20) A method disclosed herein is used to connect an equalizer bar to a walking beam of a reciprocating pump unit. The method comprises: attaching a bearing to the walking beam, the bearing having a shaft with cylindrical ends; resting hooked sections of slots defined in brackets on the equalizer to rest against the cylindrical ends of the shaft; and positioning wedges in the slots in a gap between base sections of the slots and the cylindrical ends of the shaft.
(21) The foregoing summary is not intended to summarize each potential embodiment or every aspect of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** illustrates an example of a reciprocating rod pump system known in the art.
(2) FIG. **2**A illustrates an exploded view of an equalizer bearing assembly of the prior art.
(3) FIG. **2**B illustrates a perspective view of a prior art hook-type attachment for connecting an equalizer beam to a walking beam.
(4) FIG. **2**C illustrates a detailed view of the hook-type attachment in FIG. **2**B.
(5) FIG. **3**A illustrates a front view of an equalizer bearing assembly having a bolted clamp attachment of the prior art for connecting an equalizer bar to a walking beam.
(6) FIG. **3**B illustrates a side view of the bolted clamp attachment in FIG. **3**A.
(7) FIG. **4** illustrates an example of a reciprocating pump unit having an equalizer bearing assembly of the present disclosure.
(8) FIG. **5**A illustrates a detailed perspective view of an equalizer bearing assembly of the present disclosure connecting an equalizer bar to a walking beam.
(9) FIG. **5**B illustrates another detailed perspective view, showing isolated components of the disclosed equalizer bearing assembly.
(10) FIGS. **6**A-**6**B illustrate side views of the cam wedge seated during bolt tightening of the disclosed equalizer bearing assembly.
DETAILED DESCRIPTION OF THE DISCLOSURE
(11) FIG. **4** illustrates a reciprocating pump unit or pump jack **30** of the present disclosure used to produce fluid from a well. The pump unit **30** is similar to that disclosed above with reference to FIG. **1** so that like reference numerals are used. As before, the pump unit **30** includes pitman arms **48**, an equalizer bar **49**, and a walking beam **50**. A horsehead **52** is mounted to the forward end of the walking beam **50**.
(12) In general, the pitman arms **48** are configured to translate on the pump unit **30**, and the equalizer bar **49** is hingedly connected to the arms **48**. The walking beam **50** is mounted to pivot on the pump unit **30**, and an equalizer bearing assembly **100** of the present disclosure connects the equalizer bar **49** to the walking beam **50**.

(13) During operation, for example, the pump unit **30** is driven by a prime mover **40** mounted on a pedestal above a base **32**. Structurally, a Samson post **34** on the base **32** provides a fulcrum on which the walking beam **50** is pivotally supported by a saddle bearing assembly **35**.
(14) Output from the motor **40** is transmitted to a gearbox **42**, which provides low-speed, high-torque rotation of a crankshaft **43**. Both ends of the crankshaft **43** rotate crank arms (one crank arm is not shown in FIG. **4** and the other crank arm is not visible), which have counterbalance weights **46**. Each crank arm is pivotally connected to a pitman arm **48** by a crank pin bearing **45**. In turn, the two pitman arms **48** are connected to the equalizer bar **49**, which is pivotally connected to the rear end of the walking beam **50** by the equalizer bearing assembly **100** of the present disclosure.
(15) FIG. **5**A illustrates a detailed perspective view of an equalizer bearing assembly **100** of the present disclosure connecting an equalizer bar **49** to a walking beam **50**. The equalizer bearing assembly **100** includes brackets or hooks **110**, an equalizer shaft or pin **120**, a bearing **122**, a bearing housing **140**, and cam wedges **130**. The equalizer shaft **120** has cylindrical ends that extend beyond the bearing **122**. The bearing housing **140** is disposed on the walking beam **50** and supports the bearing **122** for the equalizer shaft **120** to rotate. The hooks or brackets **110** are disposed on the equalizer beam **49**.
(16) In this detailed view, the equalizer bar **49** is shown having the pitman arms **48** connected by hangers **49***p* on the equalizer bar **49**. The hooks **110** extend from the equalizer bar **49**. These hooks **110** can preferably be welded to the equalizer bar **49** as shown, although other forms of attachment (e.g., bolted flanges) can be used. The bearing housing **140** can be affixed to the walking beam **50** using bolts or the like. When the assembly **100** is assembled, the hooks **110** fit on free ends of the equalizer shaft **120**, and the cam wedges **130** affix in the hooks **110** under the equalizer shaft **120**.
(17) FIG. **5**B illustrates another detailed perspective view, showing isolated components of the disclosed equalizer bearing assembly **100**. The hooks **110** are shown on the equalizer beam **49**, and the equalizer shaft **120** and bearing **122** are shown in isolation without the bearing housing. Each of the hooks **110** define a slot **112** having a hooked section and a base section. The hooked section of the slot **112** is configured to rest against a respective one of the cylindrical ends of the equalizer shaft **120**. The base section opposes the hooked sections at a gap from the cylindrical end of the shaft **120**.
(18) In FIG. **5**B, one of the cam wedges **130** is shown being pivoted in place in the slot **112** so the cam wedge **130** can fit between the equalizer shaft **120** and the hook **110**. When disposed in the slot **112**, the cam wedge **130** fits in the gap between the slot's base section and the shaft's cylindrical end. The cam wedge **130** has a through-passage **135** for a bolt (not shown), which threads into a bolt hole **115** exposed in the hook's slot **112**.
(19) FIGS. **6**A-**6**B illustrate side views of a cam wedge **130** being seated during bolt tightening when the equalizer bearing assembly **100** of the present disclosure is being assembled. Again, the equalizer bearing assembly **100** has an equalizer shaft **120** that is supported on the walking beam (not shown). The equalizer shaft **120** is configured to rotate by a bearing **122**. The hook or bracket **110** is disposed on the equalizer beam **49**.
(20) As noted above, the hook **110** defines a slot **112**. In detail, the slot **112** has a hooked section **114**, a face section **116**, and a base section **118**. The hooked section **114** is circumferential and is configured to rest against the cylindrical end of the equalizer shaft **120**. The base section **118** opposes the hooked section **114** at a gap from the cylindrical end of the equalizer shaft **120**. When installed, the cam wedge **130** is disposed in the slot **112** between the base section **118** and the cylindrical end of the equalizer shaft **120**.
(21) Once installed, the cam wedge **130** affixes in the slot **112** using a fastener **137**. For example, the fastener **137** can be a bolt disposed in an opening (**135**) through the cam wedge **130** and threaded into a bolt hole **115** in the slot **112**.
(22) To connect the equalizer bar **49** to the walking beam **50** during assembly of the reciprocating pump unit (**30**), the bearing assembly **140** as shown in FIGS. **5**A-**5**B is attached to the walking

beam **50**. For example, flanges on the bearing assembly **140** can bolt to the walking beam **50**. As noted, the bearing assembly **140** has an internal bearing **122** that supports the cylindrical equalizer shaft **120** to rotate. Meanwhile, the hooks **110** are attached to the equalizer bar **49**. For example, the hooks **110** can be pre-welded on the equalizer bar **49**, but other forms of attachment, such as bolting, can be used.

(23) As shown in FIGS. **6**A-**6**B, the hooked section **114** of the hook **110** is hooked to rest against the cylindrical end of the equalizer shaft **120**. The cam wedge **130** is then positioned in the slot **112** in the gap between the base section **118** and the cylindrical end of the shaft **120**. The bolt **137** is passed through the through-hole (**135**) of the cam wedge **130** and is tightened in the bolt hole **115** of the hook **110** to affix the cam wedge **130** in the slot **112**. Cam wedges **130** are installed on both of the hooks **110**.

(24) Other conventional steps can then be performed at the appropriate points in the assembly of the pump unit (**30**). For example, the pitman arms (**48**) can be attached to the equalizer bar **49**; the wristpin bearing assemblies on the pitman arms (**48**) can be attached to the crank arms; etc.

(25) The cam wedges **130** can overcome issues noted previously should the pump unit (**30**) experience compressive loading (e.g., in the case of a polished rod hang-up, high rod part, or other type of abrupt shock loading). The cam wedges **130** positively lock the hooked portion **114** of the slot **112** to the equalizer shaft **120** by camming matching profiles against the cylindrical end of the shaft **120** and the base section **118** of the slot **112**.

(26) Looking at the cam wedge **130** in detail in FIGS. **5**A-**5**B, the cam wedge **130** defines a first edge **134** against which the cylindrical end of the shaft **120** is configured to rest. Additionally, the cam wedge **130** defines a second edge **138**, which is opposite the first edge **134** and is configured to rest against the base section **118** of the slot **112**. The first edge **134** is a concave edge, defining a cylindrical profile matching the cylindrical end of the equalizer shaft **120** that rests against it. The radius R.sub.1 of curvature for this concave edge **134** is centered at the central, rotational axis C.sub.1 of the equalizer shaft **120**. Meanwhile, the second edge **138** is a convex edge, defining a cylindrical profile with a larger radius R.sub.2 of curvature that is centered at an offset central axis C.sub.2 from that of the equalizer shaft **120**. The concave edge **134** defines a first segment S.sub.1 that is larger than a second segment S.sub.2 of the convex edge **138**. This allows the cam wedge **130** to support more surface area of the cylindrical end of the shaft **120**.

(27) During assembly, the cam wedge **130** is inserted in the slot **112** between the shaft's cylindrical end and the slot's base section **118**. The bolt **137** passes through the cam wedge **130** at the opening in a front face **132** and is threaded into the bolt hole **112** in the face section **116** of the slot **112**. As the bolt **137** is tightened, the matching concave edge **134** fits up against the cylindrical shaft **110**. The cam wedge **130** rotates into place using the offset center C.sub.2 so the convex edge **138** is pulled into the correspondingly shaped base section **118** of the hooked slot **112** to create a full contact load path.

(28) The separation between the back face **136** and face section **116** closes as the cam wedge bolt **137** is tightened, and the cam wedge **130** is pivoted toward the slot's face section **116**. Tightening the bolt **137** thereby locks the cam wedge **130** in place. A solid line of contact between the shaft **120** and the slot's base section **118** creates a path to support compressive loading should it occur.

(29) Ideally, the arrangement does not produce stress concentrations or bending in the bolt **137**. Instead, the cam wedge **130** allows compressive load from the equalizer shaft **120** on the walking beam (**50**) to pass to the equalizer bar **49**. The equalizer bearing assembly **100** can be designed to take the loading so there may be no need for inspection after high compressive loading occurs during operation of the pump unit (**30**). In the end, the equalizer bearing assembly **100** uses the proven hook-style connection, which can be quickly assembled in the field between an equalizer bar (**49**) and walking beam (**50**). Meanwhile, the cam wedge **130** enables the assembly **100** to take compressive loading.

(30) The foregoing description of preferred and other embodiments is not intended to limit or

restrict the scope or applicability of the inventive concepts conceived of by the Applicants. It will be appreciated with the benefit of the present disclosure that features described above in accordance with any embodiment or aspect of the disclosed subject matter can be utilized, either alone or in combination, with any other described feature, in any other embodiment or aspect of the disclosed subject matter.

(31) In exchange for disclosing the inventive concepts contained herein, the Applicants desire all patent rights afforded by the appended claims. Therefore, it is intended that the appended claims include all modifications and alterations to the full extent that they come within the scope of the following claims or the equivalents thereof.

Claims

1. An assembly to connect arms of a reciprocating pump unit to a walking beam of the reciprocating pump unit, the walking beam having a bearing shaft with cylindrical ends, the assembly comprising: an equalizer bar having brackets, each of the brackets defining a slot, the slot having a hooked section and having a base section, the hooked section being configured to rest against a respective one of the cylindrical ends of the bearing shaft, the base section opposing a respective one of the hooked sections at a gap from the cylindrical end of the bearing shaft, the base section defining a base concave profile; and wedges being configured to pivot into position in the slots between the base sections of the slots and the cylindrical ends of the bearing shaft, each of the wedges having first and second edges, the first edge having an edge concave profile, the edge concave profile being configured to rotate against the cylindrical end of the bearing shaft, the second edge opposing the first edge and having an edge convex profile, the edge convex profile being complementary to the base concave profile of the base section and being configured to cam against the base concave profile of the base section of the slot.

2. The assembly of claim 1, wherein the assembly comprises fasteners, each affixing a respective one of the wedges in a respective one of the slots.

3. The assembly of claim 2, wherein each of the wedges defines an opening therethrough; wherein each of the slots defines a bolt hole; and wherein each of the fasteners is a bolt disposed in the opening defined through the wedge and threaded into the bolt hole in the slot.

4. The assembly of claim 1, wherein the edge concave profile defines a first segment that is larger than a second segment of the edge convex profile.

5. The assembly of claim 1, wherein the edge concave profile defines a first radius of curvature matching the cylindrical end of the bearing shaft.

6. The assembly of claim 5, wherein the edge convex profile defines a second radius of curvature at a central axis offset from a rotational axis of the bearing shaft.

7. The assembly of claim 6, wherein the base concave profile defines the second radius of curvature at the central axis offset from the rotational axis of the bearing shaft.

8. The assembly of claim 1, wherein the assembly comprises a bearing housing configured to attach to the walking beam, the bearing housing having a bearing configured to support the bearing shaft for rotation therein.

9. The assembly of claim 1, wherein the brackets are welded to the equalizer bar.

10. An assembly to connect an equalizer bar to a walking beam of a reciprocating pump unit, the walking beam having a bearing shaft with cylindrical ends, the assembly comprising: brackets configured to attach to the equalizer bar, each of the brackets defining a slot, the slot having a hooked section and having a base section, the hooked section being configured to rest against a respective one of the cylindrical ends of the bearing shaft, the base section opposing a respective one of the hooked sections at a gap from the cylindrical end of the bearing shaft, the base section defining a base concave profile; and wedges configured to pivot into position in the slots between the base sections of the slots and the cylindrical ends of the bearing shaft, each of the wedges

having first and second edges, the first edge having an edge concave profile, the edge concave profile being configured to rotate against the cylindrical end of the bearing shaft, the second edge opposing the first edge and having an edge convex profile, the edge convex profile being complementary to the base concave profile of the base section and being configured to cam against the base concave profile of the base section of the slot.

11. The assembly of claim 10, wherein the assembly comprises bolts, each affixing a respective one of the wedges in a respective one of the slots, wherein each of the wedges defines an opening therethrough; wherein each of the slots defines a bolt hole; and wherein each of the bolts is configured to position in the opening defined through the wedge and to thread into the bolt hole in the slot.

12. The assembly of claim 10, wherein at least one of: the edge concave profile defines a first segment that is larger than a second segment of the edge convex profile; the edge concave profile defines a first radius of curvature matching the cylindrical end of the bearing shaft; the edge convex profile defines a second radius of curvature at a central axis offset from a rotational axis of the bearing shaft; and the base concave profile defines the second radius of curvature at the central axis offset from the rotational axis of the bearing shaft.

13. The assembly of claim 10, comprising a bearing housing configured to attach to the walking beam, the bearing housing having a bearing configured to support the bearing shaft for rotation therein.

14. A reciprocating pump unit, comprising: arms configured to translate on the reciprocating pump unit; an equalizer bar hingedly connected to the arms; a walking beam mounted to pivot on the reciprocating pump unit; and an assembly connecting the equalizer bar to the walking beam, the assembly comprising: a shaft having cylindrical ends; a bearing disposed on the walking beam and supporting the shaft to rotate; brackets disposed on the equalizer bar, each of the brackets defining a slot, the slot having a hooked section and having a base section, the hooked section configured to rest against a respective one of the cylindrical ends of the shaft, the base section opposing a respective one of the hooked sections at a gap from the cylindrical end of the shaft, the base section defining a base concave profile; and wedges pivoted into position in the slots between the cylindrical ends of the shaft and the base sections of the slots, each of the wedges having first and second edges, the first edge having an edge concave profile, the edge concave profile rotated against the cylindrical end of the shaft, the second edge opposing the first edge and having an edge convex profile, the edge convex profile being complementary to the base concave profile of the base section and cammed against the base concave profile of the base section of the slot.

15. The reciprocating pump unit of claim 14, wherein the assembly comprises bolts, each affixing a respective one of the wedges in a respective one of the slots, wherein each of the wedges defines an opening therethrough; wherein each of the slots defines a bolt hole; and wherein each of the bolts is configured to position in the opening defined through the wedge and to thread into the bolt hole in the slot.

16. The reciprocating pump unit of claim 14, wherein at least one of: the edge concave profile defines a first segment that is larger than a second segment of the edge convex profile; the edge concave profile defines a first radius of curvature matching the cylindrical end of the shaft; the edge convex profile defines a second radius of curvature at a central axis offset from a rotational axis of the shaft; and the base concave profile defines the second radius of curvature at the central axis offset from the rotational axis of the shaft.

17. The reciprocating pump unit of claim 14, wherein the assembly comprises a bearing housing configured to attach to the walking beam, the bearing housing having the bearing configured to support the shaft for rotation therein.

18. A method to connect an equalizer bar to a walking beam of a reciprocating pump unit, the method comprising: attaching a bearing to the walking beam, the bearing having a shaft with cylindrical ends; resting hooked sections of slots defined in brackets on the equalizer bar to rest

against the cylindrical ends of the shaft; and pivoting wedges into position in the slots in a gap between base sections of the slots and the cylindrical ends of the shaft by: rotating edge concave profiles of first edges of the wedges on the cylindrical ends of the shaft, and camming edge convex profiles of second edges of the wedges on base concave profiles of the base sections complementary to the edge convex profiles.
19. The method of claim 18, wherein pivoting the wedges into position in the slots comprises bringing inside edges of the wedges toward sides of the slots by tightening the wedges in place with bolts, each bolt extending through an opening from an outside edge of the wedge to the inside edge of the wedge and affixing to a bolt hole defined in the side of the slot.
20. The method of claim 18, wherein rotating the edge concave profiles of the first edges on the cylindrical ends of the shaft comprises moving the edge concave profiles along a radius of curvature matching the cylindrical ends of the shaft; and wherein camming the edge convex profiles on the base concave profiles comprises moving the edge convex profiles on a second radius of curvature at a central axis offset from a rotational axis of the shaft, the central axis offset away from the hooked sections of the slots.
21. The assembly of claim 1, wherein the wedges pivoted into position in the slots have a solid line of contact between the cylindrical ends and the edge concave profiles and between the edge convex profiles and the base concave profiles, the solid line of contact being configured to support compressive loading between the cylindrical ends and the base sections.
22. The reciprocating pump unit of claim 14, wherein the wedges pivoted into position in the slots have a solid line of contact between the cylindrical ends and the edge concave profiles and between the edge convex profiles and the base concave profiles, the solid line of contact being configured to support compressive loading between the cylindrical ends and the base sections.